MID TERM EXAMINATION
(CBCS)
B. Tech (CSAI/ECE AI), 2nd Semester
(June, 2022)

Subject Code: BAS 106	**Subject:** Environmental Sciences
Time : 1.5 Hours	**Maximum Marks :** 30
Note:Q. 1 is compulsory. Attempt any one question from the rest.	

Q1		**(4*2.5=10)**
	(a) Explain the sources and control of CO pollution.	
	(b) Explain the concept of atom economy with the help of a green reaction from propyne to Methyl methacrylate.	
	(c) Give the classification of fuels with suitable examples.	
	(d) Discuss the ways by which water harvesting is practiced in India?	
Q2		**(5+5=10)**
	(a) List and Explain the secondary treatment of waste water treatment?	
	(b) Discuss the sources, sink and effects of oxides of sulphur as air pollutant	
Q3		**(5+5=10)**
	(a) Write a short note on (i) Salinity (ii) Salt Water Intrusion (b) What are the major issues existing in the world associated with the Food Problems and mention it's affects on human health?	
	(b) A sample of coal containing 93% C, 6% H, 1% ash. When this coal was tested for the calorific value in the bomb calorimeter, the following data were obtained:- Weight of coal burnt = 0.92 gm, Weight of water taken = 550 gm, Water equivalent of bomb and calorimeter = 2,200 gm, Rise in temperature = 2.42 °C, Cooling correction = 0.02 °C, Fuse wire correction = 10.0 cal, Acid correction = 50.0 cal. Calculate the net and gross calorific value of the coal in cal/gm. given latent heat of condensation of steam = 587 cal/g	
Q4		**(5+5=10)**
	(a) Discuss the Chemical processes for the management of Hazardous Wastes.	
	(b) Chemists some time refer to "byproducts" rather than "waste". List one advantage of using this term. How the new synthesis of Adipic acid is is better than the conventional synthesis. Explain.	